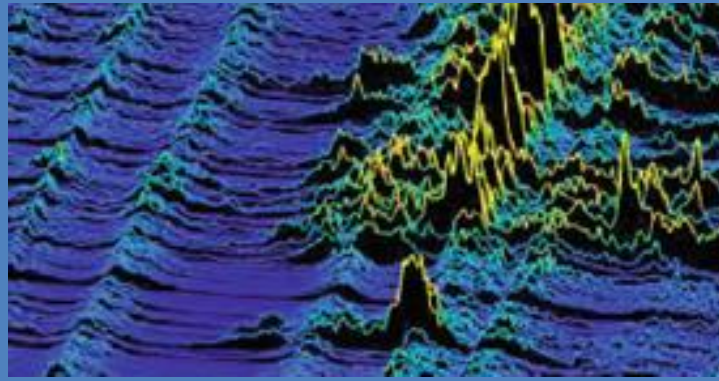




RIMA-1-420-s

Signal processing and identification in control of mechatronic devices

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This document describes information about the project, signals, exercises, grading system, and list of supplementary material.

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1 General information

According to the syllabus, the project refers to “**implementation of the identification process from the experimental works to model parameters estimation for the given case**”. Typically, students observe how data is collected with high-class equipment, and then they process it. However, it is much more beneficial for students to individually build a system, which can collect data at any time and location, form different objects, even if it is significantly less accurate and functional.

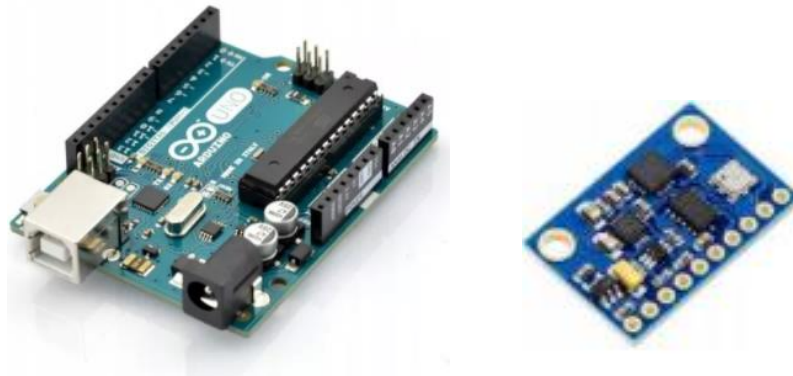


Fig. 1.1 Popular Arduino Uno® microcontroller with exemplary accelerometer

For instance, such system could be built using Arduino Uno® unit, see Fig. 1.1. However, other online and offline ways of realization of the project are also possible. For shopping, use Kamami and Botland internet stores.

2 Specification summary

Table 2.1 represents summary of work that is to be done in order to obtain particular grade from the class. Table 2.1 contains **three “MODES”**, which represent different type of data, which is going to be processed, namely: **Offline data** (provided to students), **MATLAB generated/recorded data** (either generated or recorded individually by student) and **Arduino-collected data** (collected individually by students with a measuring unit built “from scratch”).

Table 2.1 General task scheme

Final grade	MODE 1 Offline data (provided)	MODE 2 MATLAB generated/recorded data	MODE 3 Arduino-collected data
3	Signal 1 Task 1.1 - 1.3	Generation 1 Task 1.1 - 1.2	Collection 1 Task 1.1
4	Signal 1 Task 1.1 - 1.3 Signal 2 Task 2.1 - 2.3	Generation 1 Task 1.1 - 1.2 Generation 2 Task 2.1 - 2.2	Collection 1 Task 1.1 Collection 2 Task 2.1
5	Signal 1 Task 1.1 - 1.3 Signal 2 Task 2.1 - 2.3 Signal 3 Task 3.1 - 3.3	Generation 1 Task 1.1 - 1.2 Generation 2 Task 2.1 - 2.2 (or Collection 2 Task 2.1 - 2.2) Collection 3 Task 3.1 - 3.2	Collection 1 Task 1.1 Collection 2 Task 2.1 Collection 3 Task 3.1
Extra	Signal 4 Task 4.1 - 4.3	Collection 4 Task 4.1 - 4.2	Collection 4 Task 4.1
3/4/5	Accepted individually	Accepted individually	Accepted individually

As clearly observed, for each of three modes, different set of signal processing exercises is required. In general, the more effort is taken to collect the data, the less effort is required within data processing. NOTE: Presented limits refer to lower boundary, i.e. anybody can do more, and more, and more.

3 Data description

3.1 Offline data

MODE 1 refers to offline data, which is provided to students. Provided data includes a set of a few real signals collected with a professional DAQ (Data Acquisition Unit) AVM4000. Data refers to objects illustrated in Fig. 3.1-3.4 (some modifications are possible).

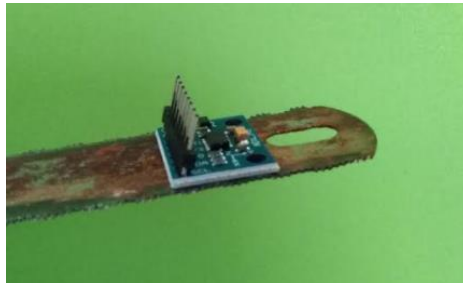


Fig. 3.1 Real vibration signal from a saw (Wood Cutter Saw works really good)



Fig. 3.2 Real vibration signal representing cyclostationary handrail vibrations



Fig. 3.3 Real vibration signal representing water heating process



Fig. 3.4 Real vibration signal from young-timer Suzuki motorcycle (GN 125)

3.2 MATLAB generated/recorded data

MODE 2 refers to data that is generated and/or recorded by student. This data is either generated synthetically in MATLAB or it is recorded in MATLAB using microphone of a personal computer (either built-in or external). Fig. 3.5 illustrates data **generation** process, which Fig. 3.6 illustrates data **collection** process in MATLAB (this scenario requires two computers).

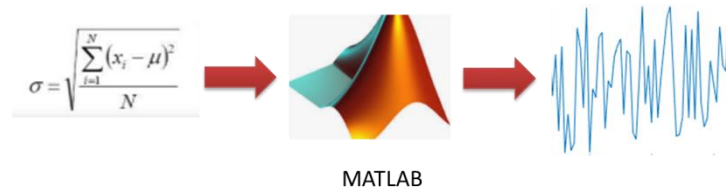


Fig. 3.5 Generation of synthetic signals using MATLAB®



Fig. 3.6 Collection of sound signals using MATLAB®

Although data for MODE 2 is conveniently generated/collected using MATLAB® scientific environment, other generation/collection methods are possible, especially using smartphone recording capabilities. Appropriate information regarding signal generation and signal recording using MATLAB® is given in SUPPLEMENTARY NOTES.

3.3 Arduino-collected data

Online real data refers to vibration, sound and/or temperature signals collected individually by a student using external measuring and data collection unit. According to APPENDIXES A-D, collected signals are generated by objects similar to ones represented offline data, namely: measurement of beam acceleration (2 types), measurement of hand rail decaying pulses, measurement of water heating process (acceleration and temperature), and measurement of IC (Internal Combustion) engine.

4 Set of signals

Appendixes A-D describe individual signals, their main characteristics, required tasks and details of project outline.

5 Grading system

Grading system is as follows:

- 1) Every student is graded according to two factors only: **knowledge** and **skills**.
- 2) Grades are given in three general scenarios:
 - following exercises from this document and presenting work individually (stationary or online),
 - following exercises from this document and sending a PDF report (see instructions at the end of each task),
 - upon individual agreement (student asks for individual project, which needs to be approved); for instance, one might focus on one task and to extra, advanced work within this task.
- 3) The list of tasks corresponding to individual grades is given in Table 2.1. This means that for lowest passing grade, one prepares a set of tasks for three different signals.
- 4) Projects are delivered at the end of class.

Generally, speaking, everyone is encouraged to learn a lot and to do a lot of work; however, it is up to students, what precisely they will do – according to their interests and needs.

6 Introductory theory

- [1] Adam Jablonski, Condition Monitoring Algorithms in MATLAB®, Springer 2021
- [2] Jérôme Antoni, Cyclostationarity by examples, Mechanical Systems and Signal Processing, Volume 23, Issue 4, May 2009, Pages 987-1036, Chapters 1 and 2

Additionally, it is recommended to read [1], section 3.2, printed hand-out, plus SUPPLEMENTARY NOTES. Additionally, it is highly recommended to get familiar with **forbot** website (best source of knowledge ever!).

A) Signal 1 – sinusoidal stationary and damped component

Table A.1 illustrates two exemplary types of “Signal 1” to be recorded and/or analyzed.

Table A.1 Description of two types of “Signal 1” - sinusoidal stationary and damped component

Provided	MATLAB-generated	Arduino-collected	Comments
signal_1_STATIONARY.mat	Open 2x MATLAB, generate (add noise) and play in one MATLAB, record in second MATLAB	 1-5 sec. no damping	Stationary deterministic (sinusoidal component) with noise
signal_1_PULSE.mat	Open 2x MATLAB, generate (add noise) and play in one MATLAB, record in second MATLAB	 10 sec. with damping	Non-stationary deterministic (single decaying pulse) with noise

Table A.2 presents a list of exercises corresponding to a particular grade.

Table A.2 List of exercises with details

Exercise	Subject	Task - STATIONARY	Task - PULSE	References
1.1	Time Waveform (TW)	read period of main deterministic component		[1] 5.7.2.2
			analyze damping parameters	
	Full-resolution spectrum	read frequencies and compare with above	read frequencies and compare with above	[1] 5.7.2.3
			compare with each other	
1.2	pwelch	analyze and compare with full-resolution spectra		MATLAB help
	spectrogram	play with “wlen”	play with “wlen”	MATLAB help
		compare with each other		
1.3	Acceleration, velocity, displacement	calculate and plot velocity and displacement waveforms		[1] 4.2.3 [1] 4.3.2 + Erratum

PDF documentation of completion of each exercise includes:

- Exercise name and number
- MATLAB code (copy-paste in small font)
- corresponding plots
- corresponding comments (**your own comments - MOST IMPORTANT PART**)
- single *printscreen* of MATLAB workspaces for MATLAB-generated data
- real photo of equipment for *Arduino-mode*
- corresponding codes for *Arduino-mode*

After this exercise, you will learn how to analyze simple deterministic stationary and simple deterministic non-stationary components.

B) Signal 2 – multiple structural pulses

Table B.1 illustrates exemplary “Signal 2” to be recorded and/or analyzed.

Table B.1 Description of “Signal 2” - multiple structural pulses

Provided	MATLAB-generated	Arduino- collected	Comments
signal_2.mat	Generate/Record using MATLAB a series of real object responses, doing repeatable, but not exactly the same “grrrrr”, “gulllll” or jingle sound	 10 sec. multiple pulses induced with some stiff object	Non-stationary random (multiple decaying pulses) with noise

Table B.2 presents a list of exercises corresponding to a particular grade.

Table B.2 List of exercises with details

Exercise	Subject	Task - MULTIPLE PULSES	References
2.1	Time Waveform (TW)	read average period of pulses	[1] 5.7.2.2
		analyze damping parameters	
2.2	Full-resolution spectrum	read dominant frequencies and compare with above	[1] 5.7.2.3
	pwelch	analyze and compare with full-resolution spectra	MATLAB help
2.3	spectrogram	play with “wlen”	MATLAB help
	User-averaged spectrum	write a code for automatic fragmentation into individual responses, compare averaging in time domain and in frequency domain; plot full-resolution spectrum; pwelch spectrum in linear scale and user-averaged spectrum on a single axis – analyze results	[1] 5.7.2.4

PDF documentation of completion of each exercise includes:

- Exercise name and number
- MATLAB code (copy-paste in small font)
- corresponding plots
- corresponding comments (**your own comments - MOST IMPORTANT PART**)
- single *printscreen* of MATLAB workspaces for MATLAB-generated data
- real photo of equipment for *Arduino-mode*
- corresponding codes for *Arduino-mode*

After this exercise, you will learn how to analyze a series of pulses of real structures.

C) Signal 3 – tea pot

Table C.1 illustrates “Signal 3” to be recorded and/or analyzed.

Table C.1 Description of “Signal 3”- tea pot

Provided	MATLAB-recorded	Arduino- collected	Comments
signal_3.mat	Record using MATLAB a relatively long signal (50-300s) which represents slow development of signal components, like power-controlled operation of a vacuum cleaner	 long non-stationary signal	place accelerometer and temperature sensor; collect data simultaneously

Table C.2 presents a list of exercises corresponding to a particular grade.

Table C.2 List of exercises with details

Exercise	Subject	Task	References
3.1	Time Waveform (TW)	plot	[1] 5.7.2.2
		calculate signal energy, power and RMS	[1] 5.5.3
		calculate signal envelope (time signal)	[1] 4.3.1
	Full-resolution spectrum	see if it makes sense	[1] 5.7.2.3
3.2	pwelch	see if it makes sense	MATLAB help
	spectrogram	play with “wlen”	MATLAB help
3.3	3-D cascade plot	input: Length (s) output: 3-D cascade plot analyze and compare with spectrogram (of the same window length)	[1] 9.5.3.3 [1] 9.5.3.5

PDF documentation of completion of each exercise includes:

- Exercise name and number
- MATLAB code (copy-paste in small font)
- corresponding plots
- corresponding comments (**your own comments - MOST IMPORTANT PART**)
- single *printscreen* of MATLAB workspaces for MATLAB-generated data
- real photo of equipment for *Arduino-mode*
- corresponding codes for *Arduino-mode*

After this exercise, you will learn how to analyze a relatively long signal, which requires preprocessing.

D) Signal 4 – IC signal

Table D.1 illustrates “Signal 4” to be recorded and/or analyzed.

Table D.1 Description of “Signal 4” – IC engine

Provided	MATLAB-recorded	Arduino- collected	Comments
signal_4.mat	Record an IC (Internal Combustion) signal using MATLAB (about 10 seconds), which represents a so-called cyclostationary signal	 cyclostationary signal	press hard and hold accelerometer by hand

Table D.2 presents a list of exercises corresponding to a particular grade.

Table D.2 List of exercises with details

Exercise	Subject	Task	References
4.1	Time Waveform (TW)	plot	[1] 5.7.2.2
		calculate signal energy, power and RMS	[1] 5.5.3
		plot Instantaneous Signal Power (use different moving windows)	
		calculate signal envelope (time signal)	[1] 4.3.1
	Full-resolution spectrum	see if it makes sense	[1] 5.7.2.3
4.2	pwelch	see if it makes sense	MATLAB help
	spectrogram	play with “wlen”	MATLAB help
4.3	understanding mechanics	using any SP techniques, analyze signals with respect to mechanical construction – for this signal use: https://www.ultimatespecs.com/motorcycles-specs/suzuki/suzuki-gn-125-1999 plus general knowledge on IC engines	

PDF documentation of completion of each exercise includes:

- Exercise name and number
- MATLAB code (copy-paste in small font)
- corresponding plots
- corresponding comments (**your own comments - MOST IMPORTANT PART**)
- single *printscreen* of MATLAB workspaces for MATLAB-generated data
- real photo of equipment for *Arduino-mode*
- corresponding codes for *Arduino-mode*

After this exercise, you will learn how to analyze cyclostationary signals.

SUPPLEMENTARY NOTES

Following code shows how to play sound and how to store it in MATLAB:

```
fs      = 2^14;                % sampling frequency
recObj  = audiorecorder(fs,16,1); % create audio objet

T      = 10;                   % signal time [s]
recordblocking(recObj, T);     % sound record starts
play(recObj);                  % play the sound (WAY 1)

x = getaudiodata(recObj);      % convert "sound" to double data
sound(x,fs);                   % play the sound (WAY 2)
```